Ancient landscapes, farming records, cities, palaces, and roads, 3200–500 BC

This dossier examines what survives for the Levant, ancient Syria, Sumer, Babylonia, and Assyria, with selected connections into Anatolia. The historical cutoff is 500 BC: later roads, later descriptions of earlier cities, and later manuscript copies are not silently substituted for contemporary evidence. Modern scholarship is used to interpret material within the period, but ancient written testimony, ancient images, excavated remains, and environmental reconstructions are identified separately.

The principal finding is that the evidence is exceptionally informative but uneven: institutions recorded agricultural work and commodities; officials reported disruptive weather and transport problems; royal inscriptions described building projects; and some tablets preserved plans or topographical lists (Metropolitan Museum irrigation tablet, Mari environmental and administrative texts, Babylonian topographical texts). Roads require particular care, because a physically surviving track, a written itinerary, a royal paving claim, and a computer-generated route are four different kinds of evidence (hollow-way archaeology, Old Assyrian route analysis).

How to read the evidence

Five evidence classes

The following labels distinguish what a claim actually rests on. They are analytical labels for this dossier, not ancient categories.

Label	Evidence	What it can establish	Principal limitation
Contemporary administrative record	Accounts, work records, delivery receipts, letters	A named institution handled a commodity, assigned work, or reported a condition; Ur III irrigation records and Samaria commodity dockets are examples (Metropolitan Museum, Ancient Hebrew Inscriptions).	An institutional account is not automatically a census of all local production or land cover.
Ancient descriptive or commemorative text	Building inscriptions, topographical lists, instructional literature	Named streets, gates, construction operations, or recommended farming techniques; examples include Nebuchadnezzar's inscriptions and the Farmer's Instructions (Babylon building inscription, CDLI P346257).	Royal self-presentation, idealization, and manuscript transmission must be assessed.
Ancient graphic or pictorial record	Maps, plans, wall paintings, reliefs	Selected spatial relationships, architectural decoration, or represented activities; Nippur's map and Mari's paintings are examples (Nippur archaeological description, Mari paintings).	A map can be selective; a painting is not an unmediated photograph.

Label	Evidence	What it can establish	Principal limitation
Archaeological feature	A street surface, gate, wall, drain, track depression, or building foundation	Physical location, material, dimensions, sequence, and associations; examples include Babylonian paving and Syrian hollow ways (Babylon paving inscriptions and finds, Brak geoarchaeology).	Dating one layer does not necessarily date a feature's first creation.
Modern reconstruction	Pollen interpretation, sediment analysis, remote sensing, geophysics, route modelling	Past vegetation, hydrological change, buried layouts, or plausible connections (Levantine pollen synthesis, Old Assyrian modelling).	The result depends on sampling, chronology, identification, and modelling assumptions.

"When" has several meanings

Every infrastructure claim should distinguish the date of the surviving record, the date of the event it describes, the first demonstrated use of the feature, and any repair or rebuilding episode. A fifth date, abandonment or infilling, may be all that archaeology supplies; at Tell Brak, pottery in hollow-way fills does not directly date the beginning of traffic (Brak geoarchaeological study).

Nebuchadnezzar's description of raising and repaving an existing processional street is therefore not a record of that route's original creation (Nebuchadnezzar II 002). Similarly, an Old Babylonian itinerary establishes a sequence of journeys or stages, not the surveyed alignment or initial construction date of every road between them (Goetze's itinerary edition).

Roads: the location-and-date catalogue

Confidence below is split conceptually between location and chronology. "High" means strong evidence for the particular claim stated, not that the entire route or its first construction is known.

Road, street, or corridor	Where it was	In-period date actually supported	Evidence and confidence	What is not established
Tell Brak radial hollow ways	Around Brak in northeastern Syria, extending from the settlement toward fields and pasture; some align with known gateways (CORONA study).	Strong third-millennium association; possible beginnings in the fourth millennium are an interpretation of formation history (Brak sections).	Physical traces, remote sensing, and sampled sections: high for surviving trace locations, broader for initial dating.	No single foundation year or demonstrated paved highway.
Northern Jazira east–west connections	Surviving segments and intermediate settlements between the regions of Tell al-Hawa, Hamoukar, and Tell Leilan, broadly along an ESE–WNW corridor (geoarchaeological reconstruction).	Networks associated with major Early Bronze Age settlements, especially the third millennium BC (same study).	Physical segments plus network reconstruction: moderate for the complete connection.	Not a continuously excavated, uniformly dated road from one city to another.

Road, street, or corridor	Where it was	In-period date actually supported	Evidence and confidence	What is not established
Tell es-Safi/Gath domestic alleyway	An excavated domestic neighborhood at Tell es-Safi/Gath in the southern Levant (Greenfield and colleagues).	Early Bronze III, approximately 2900–2400 BC in the published study (article abstract).	Architecture, stratigraphy, and associated artifacts: high for the excavated passageway.	The broad occupation range is not an exact year of first laying out the alley.
Aššur–Anatolia caravan network	From Aššur through Upper Mesopotamia and Euphrates crossings toward Kaneš/Kültepe; Hahhum–Timelkiya–Hurama is one important reconstructed chain (route study).	Early second millennium; the study dates Kültepe II approximately 1970–1835 and Ib approximately 1835–1710 BC (author's chronology).	Merchant texts and proposed geography: high for network existence, variable for intermediate locations.	No single verified continuous ground alignment or road-building date.
Old Babylonian itinerary, UIOM 2134	Textual stages include Larsa, Babylon, Sippar, northern stations, Harran, and Šubat-Enlil (text edition).	Old Babylonian; an exact regnal-year restoration proposed in the edition is conjectural (dating discussion).	Ancient itinerary with duplicate UIOM 2370: high for preserved named stages, variable for proposed identifications.	The edition's obsolete absolute chronologies and several old site identifications cannot be adopted uncritically.
Qaṭṭunān–Šubat-Enlil timber transfer	Timber was transferred from river transport at Qaṭṭunān to overland transport toward Šubat-Enlil (Mari texts ARM I 7 and 98, discussed by Reculeau).	Eighteenth-century Mari archive context (study).	Contemporary logistical letters: strong for the transport connection.	No surveyed roadbed or demonstrated date of road construction.
Arpad/Tell Rifa'at radial tracks	Around Tell Rifa'at, north of Aleppo in the Qoueiq valley, radiating from apparent lower-town gateways (remote-sensing study).	Assigned approximately 1000–700 BC, Iron Age II, through settlement associations (same study).	Physical traces and settlement morphology: high for traces, inferential for period attribution.	Not dated by an inscription naming their builder.
Neo-Assyrian King's Road network	Provincial stations connected the empire; the discussed Que–Assyria itinerary crosses the Euphrates at Til-Barsip and Tigris at Nineveh (ORACC overview).	Documented in eighth–seventh-century sources; an origin under Shalmaneser III, 858–824 BC, is proposed, not proven (ORACC).	Letters and institutional terminology: high for the system, variable for physical routes.	The overview reports no securely identified archaeological bēt mardēti station; the network's creation year is unknown.
Jordan Valley ascent routes and watchtowers	Routes rising from the Jordan Valley toward Israelite administrative areas; a tower near Poriyya is compared with towers in eastern Samaria (Be'eri and colleagues).	Iron Age II; the authors associate the system with Omride initiative (article abstract).	Excavated tower plus comparative and historical reconstruction: stronger for the structures than for the complete road-control model.	Tower date is not necessarily the date a route first came into use.

Road, street, or corridor	Where it was	In-period date actually supported	Evidence and confidence	What is not established
Nineveh royal road	Inside Nineveh; competing reconstructions connect different southern, northern, and eastern gates and the citadel zone (RINAP discussion).	Sennacherib's inscriptions composed approximately 693–691 BC describe a 52-cubit-wide royal road (RINAP).	Royal inscriptions, paving, and topographical hypotheses: high for the recorded project, uncertain for its complete alignment.	No consensus gate-to-gate line should be presented as an established surveyed fact.
Lachish access and regional corridor	Western city gate and plaza at Lachish; the site controlled a coastal-plain-to-Hebron/Jerusalem-hills corridor (IAA excavation report).	Gate/plaza assigned to Stratum II, seventh–sixth centuries BC (IAA).	Excavated gateway: high; regional corridor: a geographical and archaeological interpretation.	The report does not date the regional road's first creation or expose its complete surface.
Babylon's Ay-ibūr-šabû	South–north processional street from the Esagil/Kasikilla area toward the Ishtar Gate, beside the royal palaces (primary inscription and commentary).	Paving and raising under Nabopolassar and Nebuchadnezzar II, spanning the late seventh and sixth centuries BC (building narrative).	Excavated paving, inscribed blocks, and topographical texts: very strong for the attested monumental phase.	Successive rebuilding does not establish the original route's foundation date.
Babylon's roads toward Kish and Dilbat	The Zababa Gate is associated with the road toward Kish; the Uraš Gate with the direction of Dilbat (Tintir commentary).	Topographical tradition assigned an earlier composition history, checked against sixth-century remains (ORACC).	Textual gate–destination relationships with archaeological checks: good for direction, incomplete for full routes.	Not a precisely surveyed countryside alignment or a dated first construction.

Tell Brak: roads as accumulated landscape wear

The Brak evidence is especially valuable because it preserves movement outside city walls, not just ceremonial infrastructure inside a capital (Brak geoarchaeology). Radial routes linked settlements to cultivated areas and pasture, while fewer traces formed longer intersettlement connections (CORONA study).

Some depressions are extraordinarily broad: the sampled Brak routes are described as generally 70–100 metres wide, with wider variation across the Khabur region (Brak study). That is the width of a cumulative worn landscape feature, not evidence for a stone-paved carriageway of that width; repeated animal and other traffic helped form the depressions (formation discussion).

The dating is instructively imperfect: the authors discuss possible fourth-millennium formation and prolonged use, whereas some pottery dates the beginning of sediment accumulation only to the mid or later third millennium (section analysis). Redeposited sherds and an unknown lag between use and infilling prevent treating a ceramic date as the birthday of the road (dating qualifications).

These traces can also help reconstruct land use: the disappearance or branching of routes has been interpreted in relation to field and pasture boundaries (landscape interpretation). Such a boundary is a modern inference from movement patterns, not an ancient cadastral line recovered in writing.

Old Assyrian caravans: a network is not a polyline

The early-second-millennium merchant evidence contains named commercial settlements and routes, but only a small minority of the relevant place names are securely attached to archaeological sites (Old Assyrian route study). This produces a characteristic imbalance: connectivity can be better established than exact geography.

Different scholars reconstruct different Aššur–Euphrates approaches, and the modelling study explicitly compares these alternatives (route comparison). Its least-cost route changes substantially depending on how the Euphrates is represented, demonstrating that an algorithmically convenient corridor is not independent proof of ancient use (modelling results).

For a defensible map, securely located nodes should be plotted as points; textually supported connections should be shown separately from hypothesized ground alignments. A speculative intermediate town should not acquire apparent certainty simply because a neat road line passes through it.

The Old Babylonian itinerary: an unusually concrete travel record

UIOM 2134 preserves a substantial itinerary, with UIOM 2370 duplicating part of it; its colophon gives a total of six months and fourteen days until the traveller's return (Goetze 1953). Its stages include southern Babylonian cities and a northern journey, making it valuable evidence for travel sequence and elapsed time (edition and route discussion).

The edition's proposed restoration to Samsu-iluna year 28 is explicitly conjectural, and its alternative absolute BC dates reflect old chronological schemes (dating argument). Several geographical identifications also belong to an earlier stage of scholarship, so the safest use here is the preserved itinerary itself rather than wholesale adoption of the publication's reconstructed map (Goetze's identifications).

A total journey time should not be divided mechanically by modern straight-line distances to produce a universal ancient daily speed. Stops, damaged sections, administrative business, river crossings, and uncertain place identifications all belong in the uncertainty budget.

Assyrian imperial communications: roads as an institution

The Neo-Assyrian system had terminology for the King's Road, hūl šarri; a staging station, bēt mardēti; and relay movement, kalliu (Assyrian communications overview). SAA 1 177 concerns two stations in Ṣubutu during Sargon's reign, while SAA 18 192 attests a station at Nippur (text references).

The overview's Que-to-Assyria example specifies the Euphrates ferry at Til-Barsip, Tigris ferry at Nineveh, a Balih ford south of Harran, and Habur tributary crossings between Guzana and

Naṣibina (itinerary reconstruction). These are more useful geographic anchors than a generalized line labelled “royal road.”

The proposed four-to-five-day delivery over that example is a modern estimate, not a surviving stopwatch record (ORACC calculation). Likewise, the suggestion that the system originated under Shalmaneser III must remain a historical hypothesis rather than a securely dated founding event (origin discussion).

Nineveh versus Babylon: two different levels of spatial certainty

Sennacherib's Nineveh inscriptions give a royal-road width of 52 cubits and describe boundary stelae intended to prevent encroachment (RINAP). The same scholarly discussion presents competing alignments involving the Aššur, Sîn, Nergal, and Šamaš gates, so a precise full route remains unresolved (alternative reconstructions).

Babylon supplies a tighter convergence of ancient names, royal descriptions, and physical paving (processional-way documentation). Nbk. 5 and 6 occur on limestone paving blocks, while Nbk. 7 occurs on breccia slabs; the excavated arrangement and inscriptions identify different components of the monumental street (paving texts).

The modern reconstruction describes approximately 1,200 metres of south–north processional street, including a northern section beside the palaces and through the Ishtar Gate (street reconstruction). Nebuchadnezzar II 002, column v, distinguishes his father's prior paving, his own raising and paving of the street, and the rebuilding of gates made too low by those changes (primary building narrative).

This is unusually strong evidence for urban maintenance over time. It still does not establish when the underlying route first existed.

Levantine routes: avoid retrospective precision

At Lachish, the gate and plaza are physical archaeological features; the connection between the coastal plain and the Hebron/Jerusalem hills is a broader corridor controlled by the site (IAA report). The excavation report dates the Stratum II gate complex to the seventh–sixth centuries BC, but does not provide an equivalent construction date for the regional road (IAA chronology).

The Jordan Valley watchtower study offers a different kind of route evidence: structures interpreted as observing crossings and ascent routes toward the Israelite highlands (Be'eri and colleagues). Its proposed Omride organization is an interpretation to test, not an inscription found in a road surface naming the commissioner (article abstract).

For this dossier, familiar modern route labels are not treated as proof of an ancient official road name. The preferred description is a named endpoint pair, a valley or pass corridor, an excavated feature, and an explicitly qualified date.

Weather: what was actually written down?

Mari's letters: rainfall, soaked barley, and missed flood warnings

The eighteenth-century Mari material is among the most useful evidence because environmental conditions appear inside operational correspondence, rather than only in retrospective disaster stories (Reculeau's Khabur study). Its letters distinguish conditions at particular places and connect rainfall or flood risk to immediate administrative action (same study).

- **ARM XXVII 105, lines 5–12:** After a long period without rain at Dūr-Zabîm, rain had fallen for a month and barley had grown; the passage also reports soaked barley at Dūr-Zabîm and Dēr (text discussion).
- **ARM XXVII 106, lines 3–6:** A damaged passage describes barley at Dūr-Zabîm being soaked by the Storm-God, combining religious language with a report about crop condition (quotation and discussion).
- **A.2175, lines 12–18:** The governor of Saggarātum complains that the governor of Qaṭṭunān twice failed to announce a Khabur flood; rain had not fallen at Saggarātum, while rainfall elsewhere was distant (letter excerpt).
- **ARM XXVII 101–102:** These letters concern the danger that flooding might reach winter crops if harvesting was not completed in time (harvest and flood discussion).

These are geographically situated observations and warnings, but they do not provide modern rainfall totals or a continuous climatological station series (Mari evidence reviewed by Reculeau). A report about flood risk should also remain distinct from proof that a particular field was actually inundated.

The most revealing inference is institutional: upstream communication mattered because local weather did not fully predict conditions arriving downriver. The missed-warning complaint is evidence for an expected communication practice, not proof of a standardized empire-wide hydrological service.

Snow and ice in Assyrian plant-transport correspondence

A letter cited as SAA 5 105 reports “much snow and ice” and explains that saplings could not yet be uprooted and sent to Dur-Sharrukin (Khorsabad plantlife dossier). This is direct evidence of weather disrupting a planned operation, but the presentation does not supply an exact observation day, temperature, snowfall depth, or securely specified collection locality (same dossier).

The operational context makes the text unusually useful: it links weather, tree handling, seasonal timing, and transport. It does not establish that the same weather conditions prevailed at Khorsabad itself.

Astronomical diaries: repeated observations rather than modern instrumentation

Babylonian astronomical diaries include reports of cloud, mist, rain, severe cold, and atmospheric or celestial appearances alongside astronomical observations (Stephenson, Willis, and Hallinan). Only the early portion of that long tradition falls before the present cutoff; later diary entries are not used here to describe events before 500 BC.

VAT 4956 covers Nebuchadnezzar II's thirty-seventh year, April 568 to April 567 BC in the chronology discussed by the study (diary dating). It includes a dated report of a red glow on the night corresponding to 12/13 March 567 BC, interpreted by the modern authors as a possible auroral observation rather than explicitly identified that way by the ancient scribe (text and interpretation).

That example demonstrates a high level of chronological specificity, not a temperature or precipitation measurement. The underlying category “red glow” and its interpretation must remain separate.

Conditional omens are not automatically observations

The astronomical study explicitly distinguishes dated diary reports from conditional omen formulations such as “if” a particular glow appears (discussion of akukutu). An omen rule can preserve ideas about possible phenomena without documenting that one occurred at a named place on a particular day.

The SAA 8 catalogue, for example, indexes a report entitled “Thunder in Ab, King Ill,” assigned 672-V (SAA 8 catalogue). A catalogue title and proposed document date alone should not be transformed into a claim that a particular thunderstorm was observed on that date.

For extracting weather data, the minimum questions are whether the text reports an occurrence or quotes a rule, whether the location is explicit, and whether the date belongs to the observation or merely the document. Without those distinctions, an impressive-looking weather database can mix fundamentally different evidence.

Farming: the richest ordinary record base

The earliest end of the range: goods and rations

At Uruk, the earliest administrative writing recorded management of goods and allocation of workers' rations; a surviving proto-cuneiform barley-distribution tablet is dated approximately 3100–2900 BC (Metropolitan Museum, Uruk). This reaches close to the requested 3200 BC horizon, but the example is an institutional commodity account, not a seasonal diary describing rainfall or the appearance of fields.

The difference matters because accounting can establish organized cereal handling without identifying where every grain shipment was grown. Production, storage, redistribution, and consumption should not be collapsed into one category.

Ur III: irrigation work and field calculations

Metropolitan Museum object 56.81.51 is identified as a record of irrigation work, dated approximately 2042 BC; its possible Umma provenance is marked uncertain in the catalogue (museum record). The public entry does not provide a full translation, so it cannot responsibly support invented details about canal length, worker totals, or days worked.

CDLI P201747, British Museum BM 104801, is an Ur III field-plan and calculation tablet attributed to Umma, with field areas and sowing quantities (CDLI record and transliteration). Its combination of diagram and numerical entries is particularly valuable for connecting land geometry to agricultural management (P201747).

Together these examples establish two different record types: labor associated with water management, and spatial–quantitative accounting for cultivation. Neither should be described as a comprehensive landscape map.

The Farmer's Instructions: technical knowledge, not a farm diary

A Sumerian witness from Ur, CDLI P346257, is catalogued as Old Babylonian, approximately 1900–1600 BC (tablet record). Its manuscript date must be distinguished from any proposed earlier history of the composition.

The text instructs its audience about inspecting dikes and ditches, controlling irrigation depth, preparing fields, sowing, timing watering, harvesting, threshing, and handling equipment (translation and transliteration). Its surviving witness includes practical measures involving furrow spacing and seed placement, although several passages are damaged or uncertain (P346257).

The document is best understood here as prescriptive agricultural literature. “Water the crop at a particular stage” is evidence for recommended practice, not proof that an identified farmer watered an identified field on a known date.

Ebla: orchards, oil, cereals, and institutional consumption

The Palace G archive and associated archaeological evidence illuminate Ebla's food economy in approximately 2450–2300 BC (Ebla food-production study). The texts mention olive groves associated with the king, queen, and high officials, village oil deliveries, cereal accounting, and monthly food-ration records in the Small Archive L.2712 (same study).

The study reports one annual delivery total of 5,635 jars of oil, but its discussion does not provide a tablet identifier or a year for that figure (Ebla oil discussion). It is therefore less suitable as a reproducible quantitative anchor than an individually identified text, and “jars” should not be converted into litres without establishing the relevant capacity.

Archaeobotanical finds and processing installations supply an independent check on textual interpretation, including evidence for cereals, olives, grapes, and food processing within the palace complex (archaeological evidence). A delivery account and a physical processing area answer related but different questions: what the institution received, and what activity occurred in a particular excavated space.

Mari: crop acreage, harvesting obligations, canals, and locusts

ARM XXVII 100 records successive increases in palace cultivation at Qaṭṭunān and a shortage of labor that left land unharvested (Reculeau). ARM XXVII 37–38 concerns subsequent cultivation and disputes over harvest obligations, while ARM XXVII 28–29 describes an unsuccessful attempt to use water in the Šadîtum canal against locusts (agricultural correspondence).

These letters are especially useful for reconstructing the difference between cultivated area and successfully harvested area. They also show why a crop shortfall should not automatically be classified as drought: labor availability and pests could be decisive.

Canal maintenance is attested in ARM XXVII 25, 39–40, and 103–104, while a year-name records the “opening” of a named canal between two settlements (canal discussion). The study cautions that opening need not mean first excavation, an important parallel to the distinction between repaving and creating a road (same discussion).

Southern Levant: seasonal schedules and commodity dockets

The Gezer Calendar is a short agricultural schedule on limestone, commonly dated approximately to the tenth century BC, listing periods associated with gathering, sowing, flax, barley, vine work, and summer fruit (Brill's discussion and text). The cited discussion describes its cultural affiliation as Israelite or Canaanite, so it should not be assigned to an uncontested national bureaucracy merely because of its conventional name (same discussion).

It records a recurring agricultural pattern, not an observed year of rainfall or a village's actual yields. Modern month-by-month translations also involve interpretation and should not be mistaken for month names explicitly written in the short ancient text.

The early-eighth-century Samaria ostraca record incoming wine and oil with dates, places, and recipients; examples include ostracon 14, a jar of aged wine from Gat Pir'an, and ostracon 21, refined oil from Tawil (Ancient Hebrew Inscriptions). These are strong evidence for organized commodity movement from the countryside to the capital, but they are not orchard inventories or road maps.

Judean evidence includes administrative and military ostraca from Arad and Lachish, with the Lachish letters found in the final Stratum II context destroyed in 587/6 BC (inscriptional survey). Such records make named movements and institutional networks visible, but a dispatch between places does not reveal the exact path a messenger followed.

Salinity: an ancient observation, but not a single-cause historical explanation

Ancient texts classified some land as ki-mun, “saline ground,” from approximately 2500 BC onward, and surveys could identify parcels left uncultivated because of salinity (environmental-dynamics review). This is genuine written evidence about soil condition, not merely a modern deduction from ruined settlements.

The larger claim that salinization drove a shift from wheat to barley and then civilizational collapse is disputed; the review contrasts Jacobsen's interpretation with Powell's criticisms and alternative explanations for barley preference (historiographical discussion). The secure local

observation, a field described as saline, should not be inflated into a uniform regional trajectory.

Trees and tree cover: inventories are not forests

What written records directly reveal

Three categories are particularly informative: managed orchards, timber logistics, and royal planting projects. Each illuminates economically selected trees rather than a complete sample of vegetation.

- **Ebla's olive groves:** Administrative references connect groves, institutional ownership, villages, and oil deliveries in the third millennium BC (Ebla study).
- **Mari's timber transfers:** ARM I 7 and 98 document a river-to-overland transfer through Qaṭṭunān toward Šubat-Enlil, linking wood procurement to transport organization (Reculeau).
- **Sargon's sapling orders:** SAA 1 226 names almond, quince, and plum saplings and reports 1,000 bundles of apple trees from Laqe in the planting operation for Dur-Sharrukin (Khorsabad plantlife).
- **Nineveh's royal landscape:** Sennacherib's inscriptions describe a garden as a likeness of Mount Amanus and claim extensive waterworks associated with cultivation and managed landscapes (RINAP building activities).

“A thousand bundles” is a shipment expression, not a measured forest area. Nor does a royal demand for trees prove that the receiving city's surrounding countryside naturally supported that exact species mixture.

What modern environmental evidence adds

For the southern Levant, pollen sequences from Birkat Ram, the Sea of Galilee, Ein Feshkha, and Ze'elim provide a north–south comparison of vegetation and environmental change (pollen synthesis). The sequence's date models and sampling intervals are central to its interpretation: some samples represent decades, not single seasons (chronological methods).

The cited synthesis interprets a strong olive-pollen decline around 3000 BC as more likely linked to economic or geopolitical change than simple drying, because relatively humid conditions continued (Early Bronze interpretation). It also reconstructs a dry interval around 1250 through the end of the twelfth century BC and later increases in arboreal pollen (Late Bronze–Iron Age sequence).

Those are interpretations of particular regional archives, not proof that every part of Syria, Judah, or Mesopotamia experienced identical weather at identical dates. Changes in cultivation, political disruption, grazing, and precipitation can produce overlapping vegetation signals.

In the Brak region, charred plant remains and alluvial evidence support a reconstruction in which oak park woodland was significant during the third millennium BC (Brak environmental

study). Around Nippur, plant microremains support date-palm cultivation from the twenty-sixth century BC onward and possible changes in the balance between palms and cereals (southern Mesopotamian environmental study).

Limits on numerical tree-cover claims

An olive-pollen percentage is not automatically the percentage of land covered by olive trees. The cited Levantine synthesis uses pollen proportions to interpret vegetation and horticultural change, not as a directly measured canopy-cover map (pollen evidence and interpretation).

Likewise, tree shipments, oil receipts, royal garden descriptions, and timber use are biased toward valuable resources. They can establish presence, management, demand, or transport, but an exact reconstruction of regional woodland extent requires separate environmental and spatial analysis.

What cities and palaces looked like

Uruk around 3200 BC: monumental mud brick and colored cones

Uruk's monumental buildings around 3200 BC were made of mud brick and decorated with mosaics of painted clay cones embedded in their walls (Metropolitan Museum essay). This provides direct archaeological support for patterned, colored surfaces rather than an image of uniformly bare brown masonry.

The early accounts establish administrative activity, while buildings and decorative remains supply visual evidence (Uruk overview). They do not combine into a surviving street-by-street illustrated description of the whole city.

Ebla's Palace G: terraces, processing areas, and archive rooms

Palace G's architectural units occupied different levels, connected through terraces, retaining walls, and stairs (Ebla food-production study). The West Unit includes evidence interpreted as cereal milling and olive processing, making the palace an economic workplace as well as a political setting (archaeological discussion).

The archive's food records therefore belong within a physically differentiated complex. Archive room, processing area, circulation space, and monumental setting should not all be rendered as an undifferentiated “throne palace.”

Mari: circulation, painting, and missing upper floors

In its final early-second-millennium form, Mari's palace occupied approximately two hectares, with roughly 300 excavated rooms and courts counted by André Parrot (discovery and excavation). Estimates of 500–600 rooms and approximately four hectares of total floor area include unexcavated and upper-floor spaces, not that many fully recovered rooms (preservation discussion).

The architectural interpretation distinguishes entrance areas, religious space, shops and storage, official apartments, kitchens, administration, and residential sectors (palace architecture). Court 106, Room 64, and Throne Room 65 are particularly important to the ceremonial sequence (architecture and paintings).

Painted remains include the Investiture composition and processional or sacrificial imagery, while the arrangement of some fragmentary large compositions remains debated (painting dossier). Reconstructions should therefore distinguish preserved painted fragments from conjectural completions.

The palace also yielded approximately 15,000 complete and fragmentary tablets, including administration and royal correspondence (excavation overview). That juxtaposition of archive, building plan, and decoration makes Mari especially powerful for connecting textual operations to rooms and spaces, although the lost upper story remains a major constraint.

Nippur: a genuine ancient city plan

A clay tablet carrying a plan of Nippur is dated approximately 1250 BC in the UNESCO tentative-list description, placing it in the Kassite-period second millennium (Nippur description). The map includes walls, gates, watercourses, and major sacred or civic features, as discussed in the museum's cartographic study (Penn Museum, Ancient Cartography).

This is not a modern archaeologist's plan disguised as an ancient document. However, translating its lines into a modern georeferenced map requires checking orientation, labels, scale conventions, and archaeological fit.

The modern Nippur description also reports a major canal through the city, irrigated grain fields and orchards outside the walls, and distinct temple and residential areas (UNESCO description). Those statements are an archaeological synthesis across phases, not proof that every feature belonged to one simultaneous cityscape.

Gudea's statue: ancient planning symbolism

The Louvre's AO 2 statue depicts Gudea with a plan, stylus, and graduated ruler, and is catalogued approximately 2120–2110 BC (Louvre object record). Its inscription is a Sumerian royal votive dedication, not a palace construction notebook (same record).

The object demonstrates the representation of architectural planning as a royal activity. The catalogue does not justify treating the depicted plan as a complete blueprint of the palace where the statue was found.

Khorsabad/Dur-Sharrukin: elevated palace, sculpture, glaze, and imported trees

Khorsabad's palace stood on an artificial terrace supported by a stone wall approximately ten metres high, with a major access ramp and a triple gateway decorated with bulls, genies, and heroes (city and palace description). The walled city's quadrangular enclosure and canal supply provide a larger setting for that elevated citadel (same description).

Colorful glazed-brick decoration included floral and figural motifs; surviving fragments and early excavation records document blue, green, white, and yellow glazing (colored-brick dossier). Much once-visible decoration survives only fragmentarily, so the documentary drawings and surviving material have different evidentiary roles (preservation account).

The planting letters add another layer: the intended appearance was not solely architecture, but a managed landscape assembled through provincial procurement (plantlife dossier). A convincing reconstruction should therefore separate the archaeological building envelope, documented decoration, and textually attested planting program.

Nineveh: urban circulation joined to water engineering

Sennacherib's inscriptions describe broadening streets and squares, bridges, gardens, canals, and flood-management works (RINAP synthesis). The scholarly chronology places early diversion works around 702–700 BC and additional canal works around 700–694 BC, while distinguishing royal claims from physical identifications (dated building discussion).

The urban image is therefore inseparable from the water landscape. Yet the exact royal-road alignment and some bridge identifications remain proposed rather than certain (topographical qualifications).

Babylon: glazed gates, palaces, and a vertically changing street

Nebuchadnezzar II 002 describes the rebuilding of the Ishtar Gate with blue-glazed brick, wild bulls, and mušḫuššu dragons after street raising made earlier entrances too low (primary text, column v). The same inscription places palace works in relation to the Euphrates, walls, canals, and Ay-ibūr-šabû (palace topography).

The associated text catalogue records cedar and other woods, metal-ornamented doors, glazed architectural elements, and stone defensive construction in the North Palace account (Nebuchadnezzar building texts). These descriptions combine materials, appearance, and relationships among structures, rather than supplying a neutral architectural survey.

Tintir = Babylon supplies a complementary topographical tradition: its fifth tablet lists gates, walls, rivers, streets, and city quarters, while the larger composition glorifies Babylon and Marduk (ORACC introduction). The proposed twelfth-century compositional setting must not be confused with the date of every surviving copy or with the sixth-century archaeological city used for comparison (composition and archaeological comparison).

This distinction is especially important for the cutoff: a convenient later manuscript of an older work is not automatically contemporary evidence. The strongest reconstruction combines appropriately dated textual witnesses with independently dated excavation.

Judah: a water-construction inscription rather than a palace description

The Siloam inscription describes two teams of stonecutters meeting and water flowing from source to pool over a stated 1,200 cubits; its script is consistent with late-eighth-century BC inscriptions (University of Michigan object description and translation). It is therefore directly relevant to urban engineering, but it records a tunnel rather than a road.

The translated text does not name the commissioning king (inscription translation). Its conventional royal attribution should remain analytically separate from what the inscription itself says.

At Lachish, excavated gate rooms and paving support a reconstruction of a Stratum II fortress approximately 22 by 36 metres, approached from the southwest between towers (IAA report). This is a gate-complex reconstruction, not evidence for an equally well documented royal palace.

A chronological guide to the surviving record types

This is a selection of evidentiary milestones rather than a claim that record keeping began at every listed date. Approximate dates are deliberately retained where the sources do not warrant a year.

Date	Example	Information preserved	Essential qualification
Around **3200 BC**	Uruk monumental buildings (Metropolitan Museum).	Mud-brick architecture and painted-cone wall decoration.	Archaeological appearance, not a complete contemporary written city description.
Around **3100–2900 BC**	Uruk barley account (Metropolitan Museum).	Commodity distribution and administration.	Not a weather log or a map of all cultivated land.
Approximately **2900–2400 BC**	Tell es-Safi/Gath alleyway (Greenfield and colleagues).	Neighborhood access and passageway archaeology.	Broad phase dating, not an exact founding year.
Third millennium BC, with possible earlier formation	Brak hollow ways (geoarchaeology).	Physical movement corridors linking settlement, fields, and pasture.	Formation, use, and infilling have different chronologies.
Approximately **2450–2300 BC**	Ebla Palace G archive (food-production study).	Cereals, oil, orchards, deliveries, and rations.	Institutionally selected activity, not a regional agricultural census.
Approximately **2120–2110 BC**	Gudea AO 2 (Louvre).	Royal representation with plan, ruler, and stylus.	Not a complete palace blueprint.
Approximately **2100–2000 BC**	Ur III field plan BM 104801/P201747 (CDLI).	Areas, diagram, and sowing calculations.	Exact year not supplied in the record used here.
Approximately **2042 BC**	Irrigation-work tablet 56.81.51 (Metropolitan Museum).	Administrative record of irrigation work.	Catalogue does not supply a full translation; provenance is qualified.
Early second millennium BC	Old Assyrian merchant network (route study).	Town connections, commercial movement, and proposed stages.	Full ground alignments often unresolved.

Date	Example	Information preserved	Essential qualification
Approximately **1900–1600 BC**	Farmer's Instructions witness P346257 (CDLI).	Technical agricultural prescriptions.	Date of the manuscript, not necessarily of first composition.
Eighteenth century BC	Mari environmental correspondence (Reculeau).	Rain, crop condition, flood warnings, harvesting, canals, and timber transfer.	Local operational reports, not continuous climate statistics.
Kassite period, approximately **1250 BC** in the cited description	Nippur city-plan tablet (UNESCO description).	An ancient plan of urban features.	Map date and dates of the represented structures are not identical.
Approximately **tenth century BC**	Gezer agricultural schedule (Brill).	Recurring cultivation and harvest periods.	Approximate dating and disputed cultural classification.
Eighth century BC	Samaria ostraca (Cambridge survey).	Wine and oil deliveries, places, recipients, and regnal years.	Not an exact measure of orchard extent.
Late eighth century BC	Siloam inscription (Michigan translation).	Tunnel excavation and water flow.	Does not name a king or describe road construction.
Approximately **693–691 BC**	Sennacherib's road descriptions (RINAP).	Royal-road width and controls on encroachment.	Complete alignment disputed.
Late seventh–sixth centuries BC	Babylon processional-street works (primary inscription).	Raising, paving, gate rebuilding, and palace adjacency.	Dates improvements, not necessarily first route creation.
568–567 BC	VAT 4956 astronomical diary (astronomical study).	Dated celestial and atmospheric observations.	Ancient terminology and modern physical interpretation must remain distinct.

Technical cautions that change the historical picture

Institutional survival is a sampling bias

Much cuneiform evidence survives from palace and temple environments, and its distribution is uneven across periods and regions (environmental-hazards review). A heavily documented royal orchard can consequently be easier to reconstruct than the surrounding noninstitutional countryside.

Absence of a surviving record is not proof that an activity did not occur. Conversely, the survival of many accounts does not mean they collectively describe every field or household.

Landscape change can move the apparent problem

The Brak study reconstructs changes in the Jaghjagh's channel form and flow regime between the fourth–third millennia and the first millennium BC (hydrological reconstruction). The southern Mesopotamian study likewise emphasizes changing river and canal environments and the need to combine texts with geoarchaeology (environmental dynamics).

Modern river lines therefore cannot simply be used as the ancient baseline. A proposed crossing, harbor, irrigation intake, or road approach must be tested against the channel geometry of the relevant period.

A settlement date does not date every visible feature

The Arpad hollow ways are assigned to Iron Age II from their relationship to the settlement and apparent gateways (remote-sensing study). That is informative contextual dating, but weaker than a road surface sealed under a securely dated building or an in-situ inscription explicitly recording its construction.

Geophysical maps present a related issue: at Kurd Qaburstan, magnetometry has revealed buried urban features including streets and large buildings, but the university account does not date each detected feature individually (University of Northern Colorado report). A mapped anomaly, its interpretation as a building, and its precise chronological phase should be stored as separate claims.

No undifferentiated “proto-Judean” record tradition across the whole range

The southern Levantine examples assembled here include Early Bronze passageway archaeology, the much later Gezer agricultural schedule, and Iron Age Hebrew administrative and engineering inscriptions (Tell es-Safi/Gath study, Gezer discussion, Hebrew inscriptional survey). They should not be collapsed into a single ethnic archive extending back to 3200 BC.

Script, language, political authority, and the modern classification of a site are different variables. The inscriptional survey explicitly cautions against treating a script label as an automatic language identification (Ancient Hebrew Inscriptions).

Most productive next-stage investigations

The following priorities are analytical recommendations. They are designed to produce more reliable detail, not simply a larger list of names.

- **A road evidence register:** Give every segment its own entry with endpoints, ancient names, modern identifications, physical evidence, first attestation, repair episodes, abandonment evidence, and separate location/date confidence. Maintain distinct layers for excavated features, textual connections, and hypothetical corridors.
- **A Mari environmental microhistory:** Work text by text through ARM XXVII 100–106, A.2175, the canal correspondence, and ARM I 7/98, recording the actual observation, sender, recipient, locality, document date, and translation uncertainty. This would connect rain, flooding, harvesting, transport, and administration without importing a generalized climate narrative.
- **A Babylon street-history dossier:** Align Nbk. 2, 5, 6, 7, and 34 with the excavation sequence, separating each raising and paving event from the inherited street. Ancient names, block inscriptions, and archaeological sections should remain independently traceable.
- **A Brak landscape transect:** Compare hollow-way sections, gates, settlement phases, field scatters, watercourses, and plant remains along one defined corridor. Avoid assigning all

visible tracks the same age.

- **A Levantine orchard-evidence comparison:** Set Ebla's institutional grove and oil records, Samaria's commodity dockets, and dated pollen sequences beside one another without converting their unlike quantities into a single “forest cover” statistic.
- **A room-level palace reconstruction:** For Mari or Khorsabad, label every feature as preserved, described in an ancient text, recorded during excavation, or restored by analogy. Treat upper floors, roofs, color placement, and garden layouts as separate reconstruction problems.

The evidence already supports a richly differentiated picture: managed fields, orchards, royal planting projects, working palaces, monumental colored architecture, river-dependent transport, and road networks that ranged from worn rural tracks to inscribed ceremonial streets (Ebla, Khorsabad, Mari, Brak, Babylon). The limiting question is not whether records exist, but which kind of record can securely establish each detail of place, appearance, environment, and date.